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## Midterm

CSEE W3827 - Fundamentals of Computer Systems  
Spring 2024

March 5, 2024  
Prof. Rubenstein

This midterm contains 16 pages: a question 0 and 3 other questions, and totals 100 points. To get full credit you must answer all questions. **NO ELECTRONIC DEVICES PERMITTED!** The time allowed is 180 minutes (3 hours).

Please answer all questions on the exam paper. Blue books are available for scrap work only. We will be scanning exams and cannot utilize any writing in the blue books.

Some advice:

- Be sure to leave some time to work on each problem. The right answer to each problem does not require a very long answer.
- Be sure to start every problem. And take some time to think about how to set the problem up before you start writing.

0. (5 pts) Do the following **on each page** of the exam:

- (a) **CLEARLY** write your **UNI**, name and P-credit TA initials at the top **of each page** of the exam in the space provided (including this page).
- (b) Answer questions on the white paper. If you need additional white pages, please raise your hand, and remember to write on the top: UNI, Name, P-Credit TA, "CSEE 3827 Sp24 Question" and indicate the question number.
- (c) label solutions (e.g., 2a, 2b, 2c or 2a, b, c) if using the additional pages.

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1. (30 pts) In this problem, we will build circuits that take as input some  $n$ -bit quantity  $X = X_{n-1}X_{n-2} \dots X_0$  and convert between 2's-complement and signed magnitude. In building your circuit, you may use basic gates (AND, OR, NOT, XOR), as well as basic circuits such as decoders, multiplexers, encoders, shifters, adder, incrementer, etc.
  - (a) (10 pts) Draw a circuit that converts an  $n$ -bit number in 2's complement representation to its  $n$ -bit equivalent in signed magnitude.

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- (b) (10 pts) Describe overflow circuitry for the converter from part 1a (you can draw the circuitry or just explain the values of  $X$  where overflow occurs).

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- (c) (5 pts) Draw a circuit that converts an  $n$ -bit number in signed-magnitude representation to its 2's complement representation.

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- (d) (5 pts) Describe the overflow circuitry for this signed-magnitude to 2's-complement converter from part 1c (as before, you can draw the circuitry or just explain the values of  $X$  where overflow occurs).

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2. (25 pts) A coin sorter examines a coin and generates four boolean signals,  $B, R, T$  and  $H$ , where the signal is TRUE if the coin is respectively **B**ig, **R**ough-edged, **T**hick, and **H**eavy. The desired output is a 2-bit quantity,  $XY$ , which is different for each type of valid coin. The table below indicates how each type of coin sets the inputs  $B, R, T, H$ , and what the corresponding output  $XY$  should be:

|         | Inputs |   |   |   | Output |   |
|---------|--------|---|---|---|--------|---|
| Coin    | B      | R | T | H | X      | Y |
| Penny   | 0      | 0 | 0 | 0 | 0      | 0 |
| Nickel  | 1      | 0 | 1 | 0 | 0      | 1 |
| Dime    | 0      | 1 | 0 | 0 | 1      | 0 |
| Quarter | 1      | 1 | 1 | 1 | 1      | 1 |

- (a) (5 pts) Give algebraic expressions for  $X$  and  $Y$  in terms of inputs  $B, R, T, H$  when we only care that the outputs are correct for the four coin types listed (i.e., we “don’t care” what is output for other combinations.) (Hint: this can be done by drawing K-maps but is also easily done by eye-balling the table above. Maybe all inputs aren’t needed?)

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- (b) (10 pts) Suppose an additional output,  $V$  is added, which takes inputs  $B, R, T, H$  and outputs a 1 when and only when the coin inputs are **valid**, i.e., the input  $BRT H$  is one of 0000, 1010, 0100, or 1111 (for all other inputs,  $V$  should equal 0). Give a simplified expression for  $V$ .

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- (c) (10 pts) Suppose it is known that counterfeit coins take either the form  $BRT H = 0001$  or  $1100$  such that the only inputs to the system would be those representing valid coins or these two invalid coins (i.e., the only inputs can be one of  $BRT H = 0000, 0001, 0100, 1010, 1100$ , or  $1111$ ). Give the simplified expression for  $V$  when these are the only inputs that are of concern.



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3. (40 pts) NVIDIA is developing a new chip that's sure to make them a gazillion more dollars! To maximize profits, they want to bring down labor costs, which is where you come in!

NVIDIA has to build a processing system that can handle two types of jobs (programs): type 0 is a graphics job that takes a single clock cycle to complete, and type 1 is an AI job that takes 2 clock cycles to complete. The system consists of two processing units, a "Primary" and a "secondary" such that the system is capable of processing 2 jobs at the same time. Each clock cycle  $t$ , a job arrives to the system, with the input at time  $t$  indicating whether the job is of type 0 or type 1. At time  $t + 1$  the job is given to an available processor. If the "Primary" processor is free, it accepts and processes the job to completion (i.e., using clock cycle  $t + 1$  to complete the job of type 0, and clock cycles  $t + 1$  and  $t + 2$  to complete the job of type 1). If the "Primary" processor is busy (i.e., working on the second cycle of the type 1 job), the job is served by the "Secondary" processor.

Note that under this construction, every time a job arrives, at least one of the two processors will be available to serve it.

- (a) (15 pts) Suppose we wish to output a 1 during any clock cycle where the job being processed by the "Primary" processor has more time remaining than the job in the "Secondary" processor. A timeline example is given below:

|           |   |   |   |       |   |       |       |   |   |       |    |       |       |       |    |    |       |       |    |    |
|-----------|---|---|---|-------|---|-------|-------|---|---|-------|----|-------|-------|-------|----|----|-------|-------|----|----|
| cycle #   | 0 | 1 | 2 | 3     | 4 | 5     | 6     | 7 | 8 | 9     | 10 | 11    | 12    | 13    | 14 | 15 | 16    | 17    | 18 | 19 |
| input     | 0 | 0 | 1 | 0     | 1 | 1     | 0     | 0 | 1 | 0     | 1  | 1     | 1     | 0     | 0  | 1  | 1     | 0     | 1  | 0  |
| Primary   | 1 | 1 | 1 | 2 → 1 |   | 2 → 1 |       | 1 | 1 | 2 → 1 |    | 2 → 1 |       | 2 → 1 |    | 1  | 2 → 1 |       | 1  | 2  |
| Secondary | 1 | 0 | 0 | 0     | 1 | 0     | 2 → 1 |   | 0 | 0     | 1  | 0     | 2 → 1 |       | 1  | 0  | 0     | 2 → 1 |    | 0  |
| output    | 0 | 1 | 1 | 1     | 0 | 1     | 0     | 0 | 1 | 1     | 0  | 1     | 0     | 1     | 0  | 1  | 1     | 0     | 0  | 1  |

Note that:

- Each time  $t$  input corresponds to a job starting at time  $t + 1$ .
- The value shown in "Primary" and "Secondary" rows shows the remaining processing time for the job being processed.  $2 \rightarrow 1$  shows a type 1 job taking 2 cycles to complete. 0 in the "Secondary" row indicates no job is being processed by the "Secondary" processor.
- Note that the "Secondary" processor will receive a type 1 job only when the "Primary" still has one cycle left to process its primary job.

Draw a state machine that can implement this process. Please make sure your state labels clearly indicate what each state represents.

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- (b) (15 pts) Suppose we enhance the “Secondary” processor such that it can complete both job types in a single cycle, while the “Primary” processor behaves as before. The revised behavior is shown below. Note that a type 1 job being processed by the “secondary” processor completes in a single cycle, but is represented by a '2' to indicate the fact that it's a job that in the previous problem would have taken 2 clock cycles to complete:

| cycle #   | 0 | 1 | 2 | 3     | 4 | 5     | 6 | 7 | 8 | 9     | 10 | 11    | 12 | 13    | 14 | 15 | 16    | 17 | 18 | 19 |
|-----------|---|---|---|-------|---|-------|---|---|---|-------|----|-------|----|-------|----|----|-------|----|----|----|
| input     | 0 | 0 | 1 | 0     | 1 | 1     | 0 | 0 | 1 | 0     | 1  | 1     | 1  | 0     | 0  | 1  | 1     | 0  | 1  | 0  |
| Primary   | 1 | 1 | 1 | 2 → 1 |   | 2 → 1 |   | 1 | 1 | 2 → 1 |    | 2 → 1 |    | 2 → 1 |    | 1  | 2 → 1 |    | 1  | 2  |
| Secondary | 1 | 0 | 0 | 0     | 1 | 0     | 2 | 0 | 0 | 0     | 1  | 0     | 2  | 0     | 1  | 0  | 0     | 2  | 0  | 0  |
| output    | 0 | 1 | 1 | 1     | 0 | 1     | 0 | 1 | 1 | 1     | 0  | 1     | 0  | 1     | 0  | 1  | 1     | 0  | 1  | 1  |

Draw a revised state machine for this system. Hint: it can be done with fewer states than part (a). Again, please make sure your state labels clearly indicate what each state represents.

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- (c) (10 pts) Build the system for part (b) using JK flip flops. You can simply write down the boolean expressions for the output and the inputs to the JK flip-flops that you use.

The following state/excitation tables may be of assistance while designing sequential circuitry with JK Flip-flops

| $J(t)$ | $K(t)$ | $Q(t+1)$          |
|--------|--------|-------------------|
| 0      | 0      | $Q(t)$            |
| 0      | 1      | 0                 |
| 1      | 0      | 1                 |
| 1      | 1      | $\overline{Q(t)}$ |

| $Q(t)$ | $Q(t+1)$ | $J_Q(t)$ | $K_Q(t)$ |
|--------|----------|----------|----------|
| 0      | 0        | 0        | X        |
| 0      | 1        | 1        | X        |
| 1      | 0        | X        | 1        |
| 1      | 1        | X        | 0        |

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Please don't start your exam until you are told to begin... I'm doing my best here to keep you entertained until then...

